Redis

Read hot data. Analyze it with SQL.

Install & load

```
SET allow_extension_repositories = 'allowed';
CREATE EXTENSION REPOSITORY IF NOT EXISTS query_farm WITH PREFIX 'https://duckdb-extensions.query.farm';
INSTALL redis FROM query_farm;  LOAD redis FROM query_farm;
```

01 At a glance

Query Redis strings, hashes and lists from DuckDB, or snapshot operational data into local tables for analysis. Every Redis call selects a named DuckDB secret, keeping connection details separate from the data-access expression.

Redis keys → SQL functions → Local analysis

02 Useful Redis operations

`redis_get(key, secret)`

Read a string (NULL when missing); redis_set writes one.

`redis_hget(key, field, secret)`

Read one hash field; redis_hset writes and returns fields added.

`redis_hgetall(key, secret)`

Field and value rows; redis_hgetall_map returns one MAP per row.

`redis_lrange_list(key, start, stop, secret)`

A list slice as VARCHAR[]; 0, -1 is the whole list.

`redis_mget_list(keys, secret)`

Typed MGET: a VARCHAR[] of values, NULL for each missing key.

`redis_scan_page(cursor, pattern, count, secret)`

One SCAN step as STRUCT(cursor, keys); repeat until the cursor is '0'.

`redis_hscan_over_scan(pattern, fields, n, secret)`

Stream key, field and value rows from matching hashes.

`redis_command(['INCR', 'hits'], secret)`

Run any other command; arrays return as JSON, nil as NULL.

03 Keep in mind

Choose table-shaped results

Prefer the typed _list, _page and _map functions: values containing commas or empty strings stay unambiguous. The older redis_mget, redis_lrange and redis_scan return comma-separated strings. Wrap redis_get or redis_hget in coalesce for a default.

04 Name the connection

```
CREATE SECRET quickref_redis (
  TYPE redis,
  host 'localhost', port '6379',
  username 'app', password 'replace-me',
  db '0', timeout_ms '2000'
);
```

username, password, db and timeout_ms are optional. Creating the secret does not connect.

05 Read a hash as rows

REQUIRES SETUP

```
SELECT field, value
FROM redis_hgetall(
  'user:42', 'quickref_redis'
);
```

Requires the server and a Redis hash at user:42.

06 Snapshot matching hashes

REQUIRES SETUP

```
CREATE TABLE users_snapshot AS
SELECT key, field, value
FROM redis_hscan_over_scan(
  'user:*', '*', 100, 'quickref_redis'
);
SELECT key, count(*) AS fields
FROM users_snapshot
GROUP BY key;
```

Use a pattern containing only hashes. Subsequent analysis reads the local snapshot.

07 In practice

Scope scans and connections

Key listing uses SCAN, never KEYS, but redis_keys and redis_hscan_over_scan walk every page; narrow the pattern on large keyspaces. Each reply wait is bounded by redis_timeout_ms (30 s) or the secret's timeout_ms. Connections are pooled per secret, and every function refuses while enable_external_access is false.

Full docs & examples ↗ Source ↗